

Outcomes of Iodine 125 Plaque Radiotherapy after Initial Observation of Suspected Small Choroidal Melanomas

A Pilot Study

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Purpose: To determine the visual and survival outcomes for patients undergoing plaque radiotherapy of suspected small choroidal melanomas after observation for growth in a pilot study.

Design: Noncomparative interventional case series.

Participants: Forty-five patients with suspected small choroidal melanomas who were seen at the Bascom Palmer Eye Institute between 1974 and 2001 and had iodine 125 (¹²⁵I) plaque radiotherapy between 1991 and 2002.

Methods: Patients with suspected small choroidal melanomas were monitored by clinical examination, photography, and echography. When the melanoma grew or developed orange pigment, patients were treated with ¹²⁵I plaque radiotherapy. Outcomes included visual acuity, ocular complications, tumor dimensions, metastasis, and death.

Main Outcome Measure: Melanoma-specific mortality.

Results: In our pilot study, 1 patient died from metastatic melanoma. One patient developed metastatic disease and is still alive. One patient developed marginal tumor recurrence after treatment and after enucleation was performed. Globe conservation was achieved in 97.8% of patients. In patients with tumor margins >2 mm from the optic nerve, vision was preserved with less than a doubling of the visual angle (within approximately 2 Snellen lines of preoperative acuity) in 90.6% (29/32) of patients at 1 year postoperatively, 65.6% (21/32) at 2 years, and 52.2% (12/23) in the long term (4–11 years).

Conclusions: Five-year melanoma-specific mortality after ¹²⁵I plaque radiotherapy of suspected small choroidal melanomas was 3.9% (95% confidence interval, 0%–11.2%). This information can be used to counsel patients with small suspected choroidal melanomas and to compare with other treatment modalities. *Ophthalmology* 2005;112:1777–1783 © 2005 by the American Academy of Ophthalmology.

The management of suspected small choroidal melanomas is controversial. The question of whether it is safe to monitor these lesions until they demonstrate growth or other signs of malignant transformation has never been answered definitively.

Observational management has been justified on several grounds, including the concern for severe treatment-related

morbidity and the impossibility of establishing a definitive diagnosis of small choroidal melanoma clinically. This observational approach has been advocated based on reports that many suspected small choroidal melanomas do not grow. Gass reported that 69 of 116 suspected small and medium choroidal melanomas failed to grow if left untreated and followed up closely for at least 5 years.¹ Butler et al determined a 5-year actuarial rate of growth of 36% in a group of 293 indeterminate pigmented choroidal lesions.² In the Collaborative Ocular Melanoma Study (COMS) of suspected small choroidal melanomas initially managed by observation, Kaplan–Meier estimates of the proportion of tumors that grew large enough to be eligible for the COMS clinical trials were 21% by 2 years and 31% by 5 years.³ If all suspected lesions were treated, many patients would be unnecessarily exposed to treatment-related morbidities. Most recently, the American Brachytherapy Society's recommendations for brachytherapy of uveal melanomas included a statement that most suspected small uveal melanomas (<2.5 mm in height and <10 mm in largest basal

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